

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000821	Version:	5.0, CURRENT
	Status:	Effective	Effective Date:	08-Jul-2011
Title:	SM DG57-DG517. Fluidic module verification.			Page 2 of 7

DG-57 / DG-517 Service Manual

FLUID MODULE VERIFICATION

Procedure description
This procedure describes how to check the Fluid module: digital pump, pressure and vacuum pumps, valves, clot detector and load cells. Note: Document REGD-0000821_v5 replaces document REGD-0000821_v4.

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000821	Version:	5.0, CURRENT
	Status:	Effective	Effective Date:	08-Jul-2011
Title:	SM DG57-DG517. Fluidic module verification.			Page 3 of 7

EQUIPMENT

1. An instrument connected with a PC with the software installed

SET UP

1. Turn on the instrument.
2. Execute the ST software (Technical Service software).
3. Enter the Technical Service password (d+g_diana).

PROCEDURE 1 - Pressure and vacuum

Note: this procedure can be executed with **No Init Technical Service software** also.

Process

1. Access 'I/O' tab in ST software.

Picture 1

2. Assure that the bottles racors are properly connected.
3. Press the 'Cont. reading' button. (see Picture 1)
4. Press the 'Pumps' option in 'Digital outputs' section.

5. Check that 'Digital inputs' 'Pressure' deactivate before 30 seconds and 'Vacuum' before 60 seconds.
6. If not, check the bottles racors, the bottles caps and the pressure and vacuum sensors (see documents REGD-0000817 and REGD-0000818).

Expected Results

The software looks like Picture 1.

The pressure and vacuum pumps turn on.

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000821	Version:	5.0, CURRENT
	Status:	Effective	Effective Date:	08-Jul-2011
Title:	SM DG57-DG517. Fluidic module verification.			Page 4 of 7

PROCEDURE 2 - Load cells

Note: this procedure can be executed with **No Init Technical Service software** also.

Process	Expected Results
1. Access 'I/O' tab in ST software.	The software looks like Picture 1.
2. Press the 'Cont. reading' button. (see Picture 1)	---
3. Extract all bottles from the instrument.	---
4. Check that 'Waste', 'Sol. A' and 'Sol. B' from 'Analog inputs' are between 90 and 110. (See Picture 1)	---
5. Otherwise readjust Load cells potentiometers. (See document REGD-0000819)	---
6. Put an empty bottle in waste position.	'Waste' value changes.
7. Check that 'Waste' is between 350 and 550.	---
8. Put an empty bottle in Sol. A position.	'Sol. A' value changes.
9. Check that 'Sol. A' is between 350 and 550.	---
10. Put an empty bottle in Sol. B position.	'Sol. B' value changes.
11. Check that 'Sol. B' is between 350 and 550.	---
12. Put a full bottle in waste position.	'Waste' value changes.
13. Check that 'Waste' is between 1950 and 2650.	---
14. Put a full bottle in Sol. A position.	'Sol. A' value changes.
15. Check that 'Sol. A' is between 1950 and 2650.	---
16. Put a full bottle in Sol. B position.	'Sol. B' value changes.
17. Check that 'Sol. B' is between 1950 and 2650.	---

PROCEDURE 3 - ElectroValves

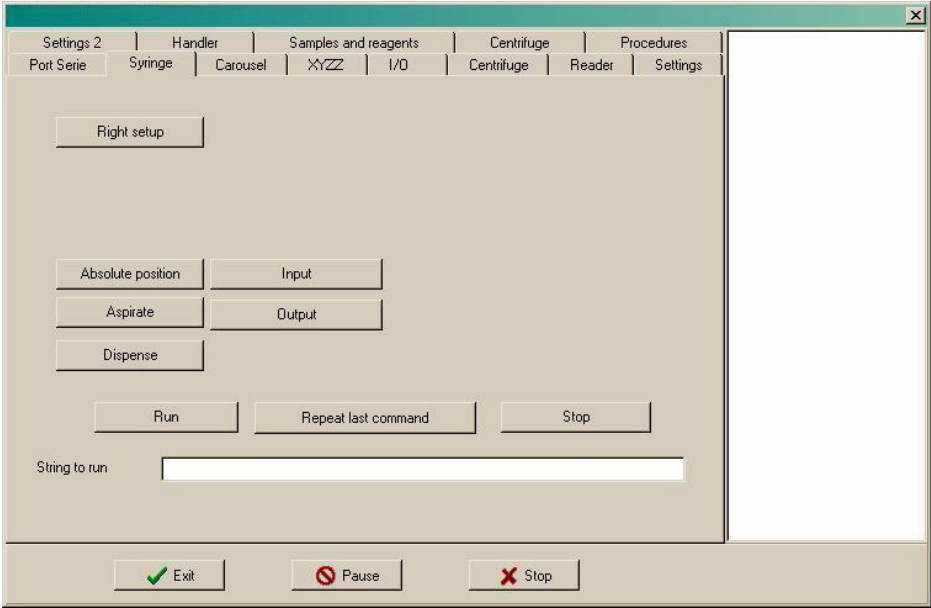
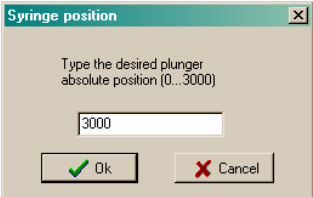
Note: this procedure can be executed with **No Init Technical Service software** also.

Process	Expected Results
1. Access 'I/O' tab in ST software.	The software looks like Picture 1.
2. Assure that the bottles racors are properly connected.	---
3. Press the 'Cont. reading' button. (see Picture 1)	---
4. Press the 'Pumps' option in 'Digital outputs' section.	The pressure and vacuum pumps turn on.
5. Wait until 'Digital inputs' 'Pressure' and 'Vacuum' deactivate.	---
6. Press the 'External wash' option in 'Digital outputs' section.	You can hear a sound indicating liquid is flowing throw the external wash station.
7. Press the 'External wash' option again.	The sound of fluid stops.
8. Press the 'Washing dilution cup' button in 'Monostables' section.	A sequence of valves activations is done.
9. Look at dilution cup and check than liquid is thrown and then aspired.	---
10. Press the 'Internal wash' option in 'Digital outputs' section.	A valve activation sound can be listened every time this option is activated.
11. Press the 'Internal wash' option again.	---

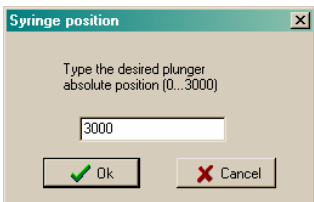
GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000821	Version:	5.0, CURRENT
	Status:	Effective	Effective Date:	08-Jul-2011
Title:	SM DG57-DG517. Fluidic module verification.			Page 5 of 7

PROCEDURE 4 - Digital Pump

Note: this procedure can be executed with **No Init Technical Service software** also.

Process	Expected Results
1. Access 'I/O' tab in ST software. 2. Assure that the bottles racors are properly connected. 3. Press the 'Cont. reading' button. (see Picture 1) 4. Press the 'Pumps' option in 'Digital outputs' section. 5. Wait until 'Digital inputs' 'Pressure' and 'Vacuum' deactivate. 6. Access 'Syringe' tab in ST software.	The software looks like Picture 1. --- --- The pressure and vacuum pumps turn on. --- The software looks like Picture 2.
 Picture 2 7. Press 'Right setup' button. 8. Press 'Input' button. 9. Press 'Aspirate' button.	--- --- The Picture 3 window appears.
 Picture 3 10 Type '3000' and press 'Ok' button. 11 Press 'Output' button. 12 Press 'Dispense' button.	--- --- The Picture 4 window appears.

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000821	Version:	5.0, CURRENT
	Status:	Effective	Effective Date:	08-Jul-2011
Title:	SM DG57-DG517. Fluidic module verification.			Page 6 of 7

Process	Expected Results
13 Type '3000' and press 'Ok' button.  Picture 4	---
14 Press 'Run' button.	The syringe starts to move.
15 Check that syringe piston moves down and then moves up.	---

PROCEDURE 5 - Clot detector

Note: this procedure can be executed with **No Init Technical Service software** also.

Process	Expected Results
1. Access 'I/O' tab in ST software.	The software looks like Picture 1.
2. Assure that the bottles racors are properly connected.	---
3. Press the 'Cont. reading' button. (see Picture 1)	---
4. Press the 'Pumps' option in 'Digital outputs' section.	The pressure and vacuum pumps turn on.
5. Wait until 'Digital inputs' 'Pressure' and 'Vacuum' deactivate.	---
6. Access 'Syringe' tab in ST software.	The software looks like Picture 2.
7. Press 'Right setup' button.	---
8. Press 'Output' button.	---
9. Press 'Aspirate' button.	The Picture 3 window appears.
10 Type '1000' and press 'Ok' button.	---
11 Press 'Dispense' button.	The Picture 4 window appears.
12 Type '1000' and press 'Ok' button.	---
13 Press 'Run' button.	The syringe starts to move.
14 Check that syringe piston moves down and then moves up.	---
15 Check that when the piston moves down the clot sensor led D2 lights on. (See document REGD-0000820).	---

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000821	Version:	5.0, CURRENT
	Status:	Effective	Effective Date:	08-Jul-2011
Title:	SM DG57-DG517. Fluidic module verification.			Page 7 of 7

DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
3.0	2005/05/09	Added Vacuum digital input deactivation time, in step 5 of procedure 1 and changed document's reference.	Document REGD-0000821_v3 replaces document 57048V0M001.
4.0	2010/09/10	Updated all the Pictures of the document.	Due to the changes introduced in last versions, new functionalities have been added to the technical service software.
5.0	See document header	Updated the Document Change History: -Date format changed to YYYY/MM/DD -Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
5.0	See document header.	Document REGD-0000821_v5 replaces document REGD-0000821_v4.	Document version changed.
5.0	See document header.	References related to manual pressure regulator deleted.	Document available for DG-57 and DG-517.
5.0	See document header.	Document title.	Commercial reference substituted by project reference.
5.0	See document header.	Document header.	Increase information to easy document search.